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BCA 5th sem

Deep Learning

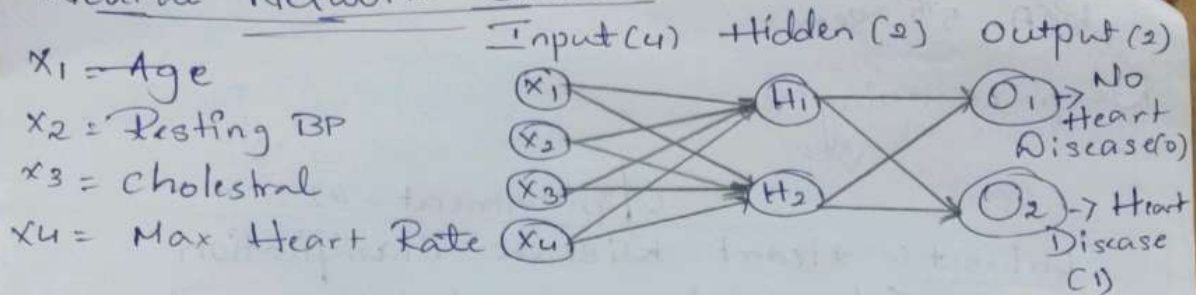
Assignment-01

Dataset:- Heart Disease Classification
(Neural Network calculation)

Problem 8- The Heart Disease dataset is a classification dataset used to predict whether a person has heart disease or not. The model uses medical features such as age, Resting blood pressure, cholesterol, and Maximum heart rate as input. A neural network with 4 input neurons, 2 hidden neurons, and 2 output neurons is used for classification. The output consists of two classes: 0 - No Heart Disease and 1 - Heart Disease. The objective is to calculate the hidden and output layer values using weights, biases, and the sigmoid activation function, and then predict the final class.

Classes:- 0 $\rightarrow$ No Heart Disease
1 $\rightarrow$ Heart Disease

Neural Network Structure



First Row Input

From Dataset (UCI Heart Disease) we take first row:

$$x = [67, 160, 286, 108]$$

Actual output: Heart Disease = 1

Input = 4

• Age = 67, Resting Blood pressure = 160, cholesterol = 286 & Maximum Heart Rate = 108

Output = 2

- No Heart Disease = (0)
- Heart Disease = (1)

Hidden Layer - Weight & Bias

For H_1 :

$$w_1 = 0.01$$

$$w_2 = 0.01$$

$$w_3 = 0.01$$

$$w_4 = 0.01$$

$$[B_1 = -0.5]$$

For H_2 :

$$w_5 = 0.01$$

$$w_6 = 0.005$$

$$w_7 = 0.005$$

$$w_8 = 0.01$$

$$[B_2 = -0.5]$$

Hidden Layer H₁

Calculate Z₁

$$\begin{aligned} Z_1 &= x_1 W_1 + x_2 W_2 + x_3 W_3 + x_4 W_4 + B_1 \\ &= (67 \times 0.01) + (160 \times 0.01) + (286 \times 0.01) + \\ &\quad (108 \times 0.01) - 0.5 \\ &= 0.67 + 1.60 + 2.86 + 1.08 - 0.5 \\ \boxed{Z_1 &= 5.71} \end{aligned}$$

Activation

$$H_1 = \frac{1}{1 + e^{-Z_1}} = \frac{1}{1 + e^{-5.71}} = \boxed{H_1 \approx 0.9967}$$

Hidden Layer H₂

Calculate Z₂

$$\begin{aligned} Z_2 &= x_1 W_5 + x_2 W_6 + x_3 W_7 + x_4 W_8 + B_2 \\ &= (67 \times 0.01) + (160 \times 0.005) + (286 \times 0.005) \\ &\quad + (108 \times 0.01) - 0.5 \\ &= 0.67 + 0.80 + 1.43 + 1.08 - 0.5 \\ \boxed{Z_2 &= 3.48} \end{aligned}$$

Activation

$$H_2 = \frac{1}{1 + e^{-Z_2}} = \frac{1}{1 + e^{-3.48}} = \boxed{H_2 \approx 0.9701}$$

Output layer - weight & Bias

For No Heart Disease (0):

$$\text{let } W_{H_1, O_1} = 0.2$$

$$W_{H_2, O_1} = 0.4$$

$$B_{O_1} = 0.1$$

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For Heart Disease (O_2):

$$\text{let } w_{h_1, O_2} = 0.5$$

$$w_{h_2, O_2} = 0.4$$

$$b_{O_2} = -0.1$$

For No Heart Disease (O_1)

Calculate z_{O_1}

$$z_{O_1} = H_1 (0.2) + H_2 (0.1) + 0.1$$

$$= (0.9967 \times 0.2) + (0.9701 \times 0.1) + 0.1$$

$$= 0.19934 + 0.09701 + 0.1$$

$$\boxed{z_{O_1} = 0.39635}$$

For Heart Disease (O_2)

$$z_{O_2} = H_1 (0.5) + H_2 (0.4) - 0.1$$

$$= (0.9967 \times 0.5) + (0.9701 \times 0.4) - 0.1$$

$$= 0.49835 + 0.38804 - 0.1$$

$$\boxed{z_{O_2} = 0.78639}$$

Final prediction

Compare the two output values:

Output	Value
No Heart Disease	0.5978
Heart Disease	0.6871

Since $0.6871 > 0.5978$

predict class = 1

Heart Disease.

Actual

class = 1 $\rightarrow$ Heart
Disease prediction
is correct.